

Albany NanoTech's Potential to Support the National Semiconductor Technology Center

By Charles Wessner and Thomas Howell

The CHIPS and Science Act of 2022 (CHIPS Act) was critically important for halting and reversing the erosion of U.S. semiconductor manufacturing capability and ensuring the security of the domestic chip supply chain. Through a combination of federal grants (up to \$39 billion) and tax incentives (in excess of \$46 billion), the CHIPS Act has stimulated billions of dollars in private sector investments into U.S.-based semiconductor manufacturing.

The CHIPS Act **established** the National Semiconductor Technology Center (NSTC), an industry-government-academic collaboration meant to drive advanced semiconductor research, development, and workforce training. In July 2024, the Department of Commerce and the recently created National Center for the Advancement of Semiconductor Technology (Natcast), the nonprofit operator of the NSTC, began the **site-selection processes** to identify the first three NSTC flagship facilities:

- An administrative and chip design center.
- An extreme ultraviolet light (EUV) lithography center will provide participants with the ability to utilize the most advanced lithography tools to facilitate the development of chips at the leading-edge of semiconductor technology, specifically those needed to support artificial intelligence applications.
- A prototyping and advanced packaging piloting facility will combine state-of-the-art 300-millimeter (mm) manufacturing with chip prototyping and packaging, set to be operational in 2028.

The Commerce Department and Natcast conducted separate site selection processes for each facility. Selection criteria **included** the presence of a thriving semiconductor ecosystem at the proposed sites, with secondary considerations including chip industry presence; the availability of skilled workers; the number and quality of relevant worker training programs; the number and quality of nearby university research programs focusing on microelectronics; the amount of public and private chip investments in the preceding decade; and the number and type of benefits, incentives, and initiatives aimed at the semiconductor industry.

In July 2024, in a bid by the state to secure the designation of the facility as one or more of the NSTC's foundational centers, \$1 billion was invested into the semiconductor research facility at the New York Center for Research, Economic Advancement, Technology, Engineering, and Science's (NY CREATES) Albany NanoTech Complex—hereafter referred to as Albany NanoTech.

Indeed, reflecting its unique advantages, in October 2024, the Department of Commerce and Natcast **designated** Albany NanoTech as the first NSTC research and development facility. Specifically, it will serve as the NSTC's EUV lithography center, supported by a proposed federal investment of an estimated \$825 million. The center is expected to begin initial operations in 2025.

In the following months, the Department of Commerce and Natcast also **announced** that the Design and Collaboration Facility would be established in California, while the Prototyping and National Advanced Packaging Manufacturing Program (NAPMP) Advanced Packaging Piloting Facility would be established at Arizona State University.

The selection of Albany NanoTech as the site for the NSTC's EUV lithography R&D center validates New York's investments in Albany. It was not lost on observers that Albany NanoTech was the first of the three NSTC flagship facility sites designated by the U.S. government, with Silicon Valley coming in second, **signaling** “a major shift in the center of power in the U.S. computer chip industry.” David Anderson, the president of NY CREATES, **observed** that the federal action was “a recognition and solidification of the Albany NanoTech center and NY CREATES as a national resource. We really are a gem for this industry and the capabilities that we have here.”

NY CREATES Albany NanoTech Complex

While the perspectives offered by Anderson might be brushed off as the sort of local boosterism commonly associated with site selections for major federal projects, his case for Albany NanoTech is grounded in a number of fundamental realities. While not fully appreciated, especially in Washington, Albany NanoTech stands at the epicenter of a long-standing, thriving semiconductor research and manufacturing ecosystem with few parallels in the United States, or worldwide. Albany NanoTech brings together cutting-edge facilities, actively engaged partners from leading device manufacturers and equipment suppliers, collaborative research relationships with leading universities, and a state government already making the significant capital investments needed for such a facility to remain state-of-the-art. Key features of this remarkable institution include the following:

- **Exceptional Facilities and Tools:** Albany NanoTech offers the best—if not the only—publicly available, neutral site capable of fully supporting the proposed NSTC EUV lithography center. Indeed, besides Belgium's imec, Albany NanoTech is the only global research center scheduled to receive the newest ASML High-NA EUV machine, set to be installed in a new facility currently

under construction. Albany NanoTech is the only such recipient in North America that is not a major private firm.

- **Bipartisan State Government Support:** The New York state government’s support for the promotion of local semiconductor research and manufacturing has been unwavering for decades, involves billions of dollars in public outlays, and remains rock-solid. It is a leading priority of Governor Kathy Hochul.
- **Large-Scale Private Investments:** The public and private sector have undertaken massive investments in chip manufacturing and research in the Albany area for the past decade. Most recently, GlobalFoundries announced an \$11.6 billion expansion of its Fab 8 in Malta, New York, beginning in 2025. Additionally, the recently announced investments at Albany NanoTech **include** another \$10 billion in public and private investments.
- **Existing High-Quality Workforce:** Over 10,000 people in the Albany area are already directly employed in semiconductor research, manufacturing, supply chain, and specialty construction companies and institutions.
- **Exceptional Educational Infrastructure:** The Albany area enjoys a dense array of elite educational and training institutions—from K-12 through post-doctoral—offering relevant curricula and training to support a semiconductor workforce. These schools are among the best in the country in their respective disciplines. The College of Nanotechnology, Science, and Engineering (CNSE), located at Albany NanoTech, is arguably among the foremost schools for applied semiconductor research in the world.
- **A Vibrant Ecosystem:** Albany is the hub of a high-tech ecosystem of semiconductor research, manufacturing, supply chain, and high-tech construction companies. Known as “Tech Valley,” the region evidences decades-long and highly sophisticated efforts by the local business community to foster knowledge-based economic development.

The New EUV Lithography Project

In December 2023, **Governor Hochul announced** a \$10 billion partnership to establish a next-generation semiconductor research and development center at Albany NanoTech—NanoFab Reflection, a research facility owned and operated by the nonprofit NY CREATES. The new center partners with Micron Technology, IBM, Applied Materials, Tokyo Electron, and other major semiconductor firms.

- **A Global State-of-the-Art Center:** As noted, the project will build a state-of-the-art High-NA Extreme Ultraviolet (EUV) Lithography Center using a lithography tool designed and built by the Dutch firm ASML. This machine is the most advanced lithography tool on the planet, and for the foreseeable future Albany NanoTech will be the only public High-NA EUV research center in North America. The ASML tool is said to cost some \$400 million and will be owned by NY CREATES.
- **The Most Advanced Equipment:** High-NA EUV lithography promises to enable the fabrication of chips at technology nodes below 2 nanometers (nm)—the current leading-edge—to **produce** devices powering generative AI and other advanced capabilities.

- **The New Facility:** NY CREATES is constructing NanoFab Reflection, a building with over 50,000 square feet of cleanroom space to house the ASML tool as well as the ancillary equipment to support experimental manufacturing operations. The capacity **expansion** is “part of an effort to land federal designation—and funding—as a hub under the National Semiconductor Technology Center.” This new facility should provide ample space for additional corporate partners and future growth.
- **Large-Scale State and Private Investment:** Few states have the wherewithal and the political will to make investments on the scale these cutting-edge facilities require. New York State **is investing** \$1 billion in the project, with private sector participants set to provide another \$9 billion.
- **Existing Facilities, Personnel, and Protocols:** Time is of the essence in the semiconductor industry, and being able to execute in real time would be a huge advantage for the NSTC. As one observer **put it**, “Albany NanoTech would be able to get the [NSTC] up and running overnight due to its existing infrastructure.”

Natcast and the CHIPS Office **state** with respect to the NSTC EUV center that

Since there are existing entities with EUV lithography capabilities, the facilities model envisions partnering with one of those entities to provide access to this capability for NSTC programs in a prompt and sustainable manner. . . . [We] envision this center to include a full-flow EUV or High Numerical Aperture EUV technology and to be operational in an existing facility by no later than 2026.

The new EUV lithography project is merely the most recent in a long line of major investments by the state of New York and its industrial partners in cutting-edge applied research infrastructure in semiconductor technology at Albany NanoTech. Even before the creation of the NanoFab Reflection lithography facility, Albany NanoTech **was** “the most advanced research facility of its kind.” Since 2009, it has housed the most advanced nonprofit 300 mm research and development (R&D) facility in the United States, which brings together academic researchers and semiconductor companies so that research can be transformed into commercial products and industrial processes.

A Key Advantage

Albany NanoTech’s current 300 mm manufacturing line is a wafer fabrication facility that enables companies to refine their manufacturing processes before they invest in their own fabs to produce their chips. Chipmaking equipment companies like ASML, Tokyo Electron, and Applied Materials can use the 300 mm line to test, prove, and gather performance data on their machines. There is no other nonprofit-owned, accessible facility in the United States that can provide these services. Most private companies cannot afford to operate a dedicated research and prototyping center on the scale and with the capabilities of the Albany facility. Even well-equipped research centers such as imec in Belgium and CEA-Leti in France are actively seeking to cooperate with Albany.

In addition to the 300 mm line, NY CREATES’s Albany NanoTech Complex enjoys an extensive array of clean room and other related facilities supporting semiconductor R&D. Since 1995, New York State and chip firms have jointly funded the successive establishment of seven major facilities at Albany NanoTech, building the densest concentration of semiconductor-related public research infrastructure

in the world. Already housing 135,000 square feet of Class I 300 mm clean room space, the completion of NanoFab Reflection will boost Albany NanoTech's total to 185,000 square feet. Only Belgium's imec, the foremost public semiconductor research center in Europe, has facilities at this scale.

Research Collaborations

Albany NanoTech is currently the site of an extensive array of public and private semiconductor research collaborations involving many of the leading chip firms in the world. The nonprofit NY CREATES, which owns Albany NanoTech, leases space to industrial tenants who conduct on-site R&D. A major industrial tenant is IBM, which **leases** more than 70,000 square feet of office and laboratory space.

Onsite projects are typically staffed by a combination of engineers, scientists, and secondees from participating companies, in addition to faculty and students from the adjacent CNSE or from other academic partnering institutions, such as Rensselaer Polytechnic Institute, which also has a growing presence at the site. The projects listed below highlight additional examples of collaboration:

- **Cooperation with Rapidus, IBM, and imec:** A recently established chip manufacturing start-up backed by eight major Japanese companies and the Japanese government, Rapidus aims to restore Japan's competitiveness in semiconductors. Their ambitious goal is to develop the capability to manufacture chips at the 2 nm node, the next generation of chips to enter high-volume production, in partnership with **IBM Research** as well as **Belgium's imec**, Europe's leading microelectronics center. Rapidus is engaged in a collaboration with IBM at Albany NanoTech to develop "Nanosheet" gate-all-around field-effect transistor (GAAFET) technology, which is essential to manufacturing 2 nm devices. To that end, Rapidus reportedly is **assigning** 100 engineers to the project in Albany. Notwithstanding its own exceptional capabilities, imec itself plans to send a team of engineers to collaborate with Rapidus at Albany NanoTech. It is interesting to note that most of the leading equipment and materials vendors, Europe's leading cooperative research center, as well as this major Japanese national initiative all seek to benefit from the facilities and expertise developed at Albany NanoTech.
- **The AI Hardware Center:** Launched in 2019 and housed at Albany NanoTech, the IBM Research AI Hardware Center serves as a research center for the development of hardware to support AI. IBM **committed** \$2 billion to establish this project, which focuses on device research, development, prototyping, and simulation. New York State's economic development fund, Empire State Development (ESD), **committed** to providing a \$300 million capital grant so that NY CREATES can purchase and install the tools necessary to support the AI Hardware Center's work.
- **The META Center:** An \$850 million project, the Materials Engineering Technology Accelerator (META) Center opened in 2019 and was jointly **established** by Albany NanoTech and Applied Materials to enable chipmakers to prototype new materials and devices. New York's ESD **committed** \$250 million to purchase and install tools in the new facility, and Applied Materials committed an additional \$600 million over seven years, with much of the investment supporting campus facilities in exchange for Applied Materials' use of clean rooms and equipment.

- **Center for Semiconductor Research (CSR):** Founded in 2005 at Albany NanoTech, CSR is an ongoing partnership with a number of leading companies in the semiconductor industry. With its 300 mm wafer fabrication facility, CSR **integrates** device design, fabrication, modeling, testing, and pilot manufacturing. It began with work at the 32 nm node of complementary metal-oxide semiconductor (CMOS) technology and has progressed to the 7 and 5 nm nodes. Work at CSR **includes** graphene-based devices, neuromorphic computing, and AI hardware. Participating companies have **included** IBM, Applied Materials, Tokyo Electron, and ASML.
- **TEL Technology Center America (TTCA):** Established in 2003, TTCA is the flagship R&D headquarters of Tokyo Electron Limited (TEL), one of the leading makers of semiconductor manufacturing equipment in the world. When the project was launched TEL **committed** \$200 million over seven years, and the state of New York committed another \$100 million. Currently, TTCA **utilizes** “16,000 square feet of clean room space [at Albany NanoTech] and 80 TEL tools with full flow integrated processing and patterning capabilities.”
- **AIM Photonics:** AIM Photonics is a federally funded, unincorporated research center based at Albany NanoTech and focusing on technology for the manufacturing of integrated photonic devices. **Created** as a federally supported manufacturing institute through an agreement between the U.S. government and the SUNY Research Foundation, AIM works with companies, universities, and government organizations to support design, research and development, proof of concept, prototyping, validation, and transfer to manufacturing of integrated photonic devices.

While research collaborations at Albany NanoTech vary, most involve multiple device and equipment firms conducting on-site research utilizing the equipment and facilities owned by NY CREATES. These arrangements enable the sharing of costs and risks involved in identifying problems and hurdles in a precompetitive, genuine manufacturing environment, thus enabling them to be addressed before companies make their own commercial investments. As a team of lithography researchers **noted** in a 2006 joint paper:

The early learning for the user in developing processes, and the feedback required by the supplier to enable improvements in the equipment technology is critical. But the cost of developing new technology is only going up, along with the risk. The best approach is through a consortium of users, equipment suppliers, universities and government. In an environment where costs and risks are shared, where the possibility exists to provide funding for equipment development and no one company bears the sole financial burden.

A Supportive, Well-Resourced State Government

NY CREATES traces its origins to a succession of state government initiatives spanning over five decades, in which a series of Republican and Democratic governors and state assembly representatives consistently supported university-based economic development centered on applying science and technology to the development of commercially relevant products and industrial processes. The initial effort was backed by massive and continued public outlays for research infrastructure and curriculum development at the state’s universities, particularly within the SUNY system.

The continuity has been remarkable. As the authors of this white paper **noted** in their 2020 study, “No governor has fundamentally reversed course or sought to erase the achievements of their predecessors. This policy continuity, often rare in state government, is an important source of the region’s success.” In her 2024 State of the State address, Governor Hochul **said** that she has “been laser-focused on developing industries of the future like semiconductors across the state.”

Phased Expansion over Three Decades

Albany NanoTech dates to 1993 when then-Governor Mario Cuomo launched the state-backed Center for Advanced Technology (CAT) at SUNY Albany, which was housed in a new 75,000 square foot facility, the Center for Environmental Sciences and Technology (CESTM). Through a series of major public and private investments aimed at building world-class facilities and acquiring cutting-edge equipment, the CESTM facility expanded in phases, a cadence that has continued to the present day. Governor Hochul’s 2023 **announcement** of the new lithography initiative, which featured a leading-edge ASML High-NA EUV lithography tool purchased by the state, is merely the most recent manifestation of the state’s ongoing commitment to invest at scale.

NY CREATES

NY CREATES is a nonprofit organization that owns and operates the Albany NanoTech Complex, where it is headquartered. **Its mission** is to “serve as a resource for public-private and academic partnerships” aimed at stimulating innovation and commercialization of technology projects. NY CREATES also facilitates efforts aimed at developing nanoelectronics, advanced analog, and digital technologies.

NY CREATES was formed in 2019 to take over the functions of two earlier nonprofit entities—Fuller Road Management Corporation and Fort Schuyler Management Corporation—that SUNY Polytechnic Institute had used for construction projects at the location and as well as at sites in Buffalo, Syracuse, Rochester, Plattsburgh, and Utica. NY CREATES’s mandate extends to the entire SUNY system, a broader role than either of the predecessor organizations, which were primarily **engaged** in contract bids and real estate acquisitions.

The original rationale for creating the Fuller Road and Fort Schuyler corporate 501(c)(3) entities under the auspices of the SUNY Research Foundation remains applicable, to a significant degree, to NY CREATES. Potential industrial partners remain more comfortable relying on a nonprofit corporate entity able to be responsive in an industry-relevant timeframe rather than working with an academic institution subject to university rules governing decisionmaking, investments, tenure, seniority, and human resources policies. Faculty members serving Albany NanoTech were employees of the two corporate nonprofit entities and thus not constrained by university rules; the same is true under NY CREATES. This use of foundations able to be responsive to industry needs proved crucial to the success of the Albany enterprise. Without this autonomy, some observers **believe** the Albany experience “never could have happened.”

Exceptional Education and Training Infrastructure

The creation of Albany’s Tech Valley has been driven by the engineering and science departments of local universities, particularly SUNY Albany’s College of Nanotechnology, Science, and Engineering (CNSE), Rensselaer Polytechnic Institute (RPI) in Troy, and the unique degree programs offered by

Union College in Schenectady. All told, the area around Albany has over 20 colleges and universities offering broad and relevant curricula, **characterized** by one local educator as:

a rich stew of higher education institutions that offer virtually anything that economic development specialists or corporate relocation specialists look at when they want to locate their plants. . . . We don't produce wafers for silicon plants, but we produce trained personnel to build those facilities.

The College of Nanotechnology, Science, and Engineering (CNSE) was formed within SUNY Albany in 2004 under the leadership of physics professor Alain Kaloyeros. CNSE operated partially within Albany NanoTech, with faculty and students participating in the complex's research activities. In a 2010 article, Professor Laura Schulz, a faculty member at CNSE, **observed** that when CNSE was established:

A new college was formed from the base up. The positions of faculty and staff were not constrained by the traditional academic expectations, but were redefined to maximize technology transfer and economic development. Faculty and staff have been hired to enable the development of ties between companies. . . . Faculty members experienced in industrial research better understand the needs of corporate partners, are able to identify potential collaborators, and can expedite the development of university and industry alliances.

CNSE currently **offers** curricula relevant to semiconductor technology, artificial intelligence, computer science, nanotechnology (including nanobiology), engineering sciences, and environmental sustainability. Undergraduate, graduate, and PhD programs are offered in nanoscale science and nanoscale engineering. The degree programs are closely tied to CNSE's R&D projects with industrial partners—most advanced-degree students take part in those projects, some even remaining as post-docs.

In 2014, Kaloyeros persuaded then governor Andrew Cuomo to spin off CNSE and merge it with the Utica-based SUNY Institute of Technology, forming a new entity—SUNY Polytechnic Institute. In 2022, Governor Hochul proposed bringing CNSE back into SUNY Albany and merging it with that institution's newly formed engineering school. That proposal was implemented in 2023, and SUNY Polytechnic presently **operates** as a separate entity from CNSE.

Rensselaer Polytechnic Institute (RPI), the oldest degree-conferring engineering school in the English-speaking world, is often characterized as “MIT on the Hudson.” It primarily focuses on providing the best-quality undergraduate engineering training, although during her tenure (1999-2022), President Shirley Ann Jackson dramatically improved graduate programs—as of 2023, four RPI graduate programs in the School of Engineering **ranked** in the top 30 nationally according to U.S. News and World Report. Her successor at RPI, Martin Schmidt, oversaw the installation of the first IBM Quantum System on a university campus. RPI, along with NY CREATES, IBM, Cornell University, and the University of Albany, also received \$40 million to create NORDTECH, a regional Microelectronics Commons Hub funded through the Department of Defense. Over the past two years, RPI has been expanding its faculty specializing in semiconductor-related research and is now offering programs jointly with local chipmakers. Recently, RPI **secured** approval for a master's program in semiconductor technology and has refurbished its cleanroom with assistance from the Department of Defense.

Union College, a private college in Schenectady, New York, has embraced what is now known as the “balanced college concept” since 1845; the curriculum exposes undergraduates both to science/engineering and the liberal arts. For generations, this formula has produced leaders in politics, science, and business, including IBM’s John E. Kelly III, a 1976 graduate who was one of the principal architects of Tech Valley.

Community colleges in the Albany area offer curricula relevant to semiconductor manufacturing, targeting the technicians and operators who make up roughly two-thirds of a chip plant’s workforce:

- **Hudson Valley Community College (HVCC)** **offers** degree programs and certifications featuring “highly specialized training in semiconductor and nanotechnology, digital electronics, electro-mechanical devices, semiconductor manufacturing and the nanofabrication process.” In 2021, HVCC **announced** a semiconductor manufacturing apprenticeship program with GlobalFoundries, giving full-time workers at the company’s fabrication plant “in-depth training and education they could almost never obtain at a liberal arts college or university.”
- **Schenectady County Community College (SCCC)** **offers** a two-year degree program in nanoscale materials technology through which students can take courses at SUNY Polytechnic and transfer into its bachelor’s program in nanoscale engineering.
- Other community colleges in the area offering degrees in semiconductor and nanotechnology **include** Mohawk Valley Community College, Fulton Montgomery Community College, and SUNY Adirondack.

New York’s K-12 schools outperform those of most other states and the schools in the Albany region collectively outperform New York state averages. NY CREATES and its industry and academic partners **conduct** numerous outreach programs in local schools with the result that most of its incoming freshmen have been coming to the Albany NanoTech Complex “since middle school.”

Tech Valley: The Albany Technology Cluster

Albany NanoTech exists in and is an integral part of a technology cluster in the Albany area comprised of mutually supporting semiconductor research, manufacturing, supply chain, and specialized construction firms. The formation of the Tech Valley cluster reflects the long-standing efforts of regional business and economic development organizations to create a center of high-tech industry in a region struggling to reverse economic decline.

Most notably, the Center for Economic Growth (CEG)—formed under the auspices of the Albany-Colonie Chamber of Commerce—functioned as a think tank and advocacy organization supporting high-tech economic growth. CEG’s efforts were strongly supported by National Grid, a local power transmission company with a long-standing commitment to local economic development, as well as local economic development organizations, including the Saratoga County Economic Development Corporation (SEDC). These organizations commissioned technical and environmental studies that helped build popular support for the Global Foundries investment and, in parallel, established a deep technical underpinning for proposals to the semiconductor industry.

The efforts and results achieved by CEG, National Grid, SEDC, and other local business groups are documented in a 2020 book published under the auspices of Georgetown University and supported by a broad range of government and private sponsors. The study, prepared by the authors of this report and entitled *Regional Renaissance: How New York's Capital Region Became a Nanotechnology Powerhouse*, describes how local leaders and key members of the state assembly developed a regional approach to economic development that was able to override parochial and competing tendencies of local governments, deploy collective pitches to companies outside the region, and leverage the Albany area's excellent K-12 schools, universities, and community colleges to attract major corporate levels of investment in semiconductor manufacturing. The accomplishments of the region offer valuable lessons for regional efforts to leverage the CHIPS Act funding.

Today the success of that effort is reflected by the presence in the region of a constellation of semiconductor and semiconductor-related companies. IBM has been a major presence in the Albany area for over half a century and currently conducts much of its semiconductor R&D at Albany NanoTech, where it is joined by TEL, Applied Materials, and now Micron. The complex and the cooperative recruiting effort attracted AMD and what became GlobalFoundries (GF), presently the only major semiconductor foundry in the United States.

Established just north of Albany in Saratoga County, GF began manufacturing operations in 2011-2012. Reflecting its importance within the national ecosystem, in February 2024, the Department of Commerce announced a \$1.5 billion award to Global Foundries pursuant to the CHIPS Act to expand and upgrade the company's manufacturing facilities. In nearby Marcy, Albany NanoTech played a major role in the decision of Wolfspeed, a semiconductor manufacturer, to build a fab in the region. More broadly, GlobalFoundries and Albany NanoTech have attracted numerous chip supply chain firms to the Albany region, demonstrating that the area is already capable of supporting manufacturing and R&D operations. The cooperative initiatives by the leading research centers in Belgium, France, Germany, and Japan underscore the quality and unique advantages of Albany NanoTech's staff and facilities.

Conclusion

When they launched the NSTC site selection process, the Department of Commerce and Natcast **indicated** that they would pick the site or sites for the first three foundational facilities "deemed the most advantageous to the objectives of [the CHIPS Act]." In other words, sites will not be picked based on political considerations—far too common in decisions for federal projects. Given today's global challenges, too much is at stake for politics to dominate capabilities.

Put simply, the Albany NanoTech Complex is a unique national asset, one that can and should serve the objectives of the CHIPS Act specifically and, more generally, the promotion of a more robust national semiconductor ecosystem able to exploit new opportunities and provide a more resilient and secure U.S. economy. Importantly, it offers an exceptional opportunity for the government to leverage existing high-cost resources in an industry-relevant timeframe.

The stakes are very real. As Governor Hochul **said** when she unveiled the EUV lithography project in 2023, speaking in the tradition of the long line of state governors who laid the foundations for today's Albany NanoTech,

This is a race for dominance, tech dominance. . . . The U.S. used to dominate this industry. . . . We're just too dependent on chips and components from places like China, Korea, [and] Taiwan. . . . So, there's massive implications for national security, technological innovation, economic growth, and independence. And the Chinese in particular are attempting to dominate this industry. . . . We have no intention of letting that happen.

This awareness of the global challenge and the willingness to devote major state resources to meeting it, combined with the existing infrastructure, make New York's Albany region an exceptional candidate for significant and ongoing support at the federal level. Successful cooperative research organizations are all characterized by sustained and substantial government support as well as active private sector participation. Timely disbursement of existing federal commitments—and proactive approaches to additional investment needs, both at the federal and state level—will be necessary to continue and reinforce the success of the nation's premier semiconductor research center. The stakes cannot be overstated. ■

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This report was made possible through general support to CSIS. No direct sponsorship contributed to this report.

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